

A Quantum Twin of the Frustrated Antiferromagnet TmMgGaO_4 on a Rydberg Atom Array

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Idea: TmMgGaO_4 is a layered magnetic crystal in which antiferromagnetically interacting moments occupy a triangular lattice. The geometry prevents all neighbouring interactions from being satisfied simultaneously. This frustration, combined with quantum fluctuations, magnetic anisotropy, disorder, temperature, and applied field, produces multiple collective magnetic regimes.

1. Pasqal and the neutral-atom platform

What makes the platform different

- The qubits are identical and free ^{87}Rb atoms. There is no fabrication variability, no yield problem and no lithography, unlike superconducting qubits, where every device must be individually made and calibrated.
- The geometry is programmable. Atoms are held in optical tweezers whose positions we choose. There is no fixed chip topology.
- Interactions are switched on by the computational state itself. Two ground-state atoms $7\text{ }\mu\text{m}$ apart do not interact at all. Excite them to a Rydberg state and they shift each other's energy by megahertz.
- Room temperature. There is no dilution refrigerator, only a vacuum chamber and lasers. Scaling from 100 to 1000 atoms is only an optics problem.

2. Analog versus digital quantum computing

Digital (gate-based)

The computation is a circuit: a discrete sequence of unitary gates drawn from a universal set,

$$|\psi_{\text{out}}\rangle = U_n \cdots U_2 U_1 |\psi_{\text{in}}\rangle.$$

The hardware Hamiltonian is hidden, and a compiler converts gates into pulses. To simulate a physical Hamiltonian H one must Trotterise e^{-iHt} into many small gate layers, at a cost that grows with accuracy and evolution time. This approach is universal, and it is the route to error correction, but it is enormously wasteful for many-body simulation.

Analog

There are no gates. You write down the Hamiltonian and let the system evolve according to it:

$$|\Psi(\Delta t)\rangle = \mathcal{T} \exp\left(-\frac{i}{\hbar} \int_0^{\Delta t} H(t) dt\right) |\Psi_0\rangle.$$

The program is $H(t)$. In Pulser it is shaped by two terms: the positions of the atoms, which give the interaction terms, and the laser amplitude, detuning and phase over time, which give the driving terms. The Hamiltonian is the centerpiece, and thereby the language.

Why analog is an optimal fit for this project

The Rydberg array's native Hamiltonian is a transverse-field Ising antiferromagnet and therefore has the same Hamiltonian that describes TmMgGaO₄ (there is a slight correction that is accounted for). The machine does not simulate a frustrated magnet by executing a program that describes one but instead acts as a frustrated magnet itself, made of a tunable atom array instead of a thulium ion lattice.

3. What physically happens inside the device

Every stage of the physical process maps onto part of the code.

Stage 1: Trapping and sorting, or the Register

A cloud of ⁸⁷Rb is laser-cooled to tens of microkelvin. A spatial light modulator splits one laser beam into hundreds of tightly focused spots acting as optical tweezers, each of which is a microscopic potential well that attracts an atom to its intensity maximum and locks it into its position.

Each tweezer holds exactly one atom, guaranteed by the mechanism of collisional blockade: if two atoms enter, a light-assisted collision ejects both. Loading is stochastic, at roughly 50 to 60 % per site. A camera images the array, a computer identifies which traps are filled, and a second set of moving tweezers rearranges the atoms one by one into a defect-free target pattern. This atom-by-atom sorting is what makes an arbitrary geometry possible. It happens before the quantum program starts. This manipulation part of the whole process is the most time-consuming and will in future be optimised.

Stage 2: State preparation

Optical pumping puts every atom in the same ground state (defined as 0). The initial state is therefore always the product state of all the atoms being in the ground state

$$|\Psi_0\rangle = |gg \dots g\rangle.$$

The goal state must be reached by evolution of the Hamiltonian.

Stage 3: Rydberg excitation

The atoms are released from this initial controlled preparation constraint and a single global laser illuminates the whole array, driving

$$|g\rangle \longleftrightarrow |r\rangle = \left|60S_{1/2}, m_j = +\frac{1}{2}\right\rangle.$$

In a Rydberg state, the valence electrons are promoted to principal quantum number $n = 60$. The electrons orbit at a radius of about $n^2 a_0 \approx 0.2 \mu\text{m}$ (with a_0 as the lattice spacing), which is a factor of ten thousand times larger than a ground-state atom. Its polarisability scales as n^7 , so two Rydberg atoms have a massive induced-dipole to induced-dipole (van der Waals) interaction proportional to n^{11}/R^6 , R being the distance between two atoms.

- Ground-state atoms $7 \mu\text{m}$ apart are non-interacting. Upon being excited, they interact strongly and shift each other's energy by megahertz.

Because the beam is global, every atom sees the same $\Omega(t)$, $\Delta(t)$ and $\phi(t)$. Shaping those three functions is the core of the programming model.

Stage 4: Readout

The optical tweezers are activated again and the array is imaged by fluorescence. The tweezers repel Rydberg atoms so these ones are lost. Therefore:

- an atom still present and glowing was in $|g\rangle$, and is recorded as bit 0;
- a dark, empty site was in $|r\rangle$, and is recorded as bit 1.

One shot yields one bitstring and is physically meaningless. Thousands of shots build the stochastic distribution we need for our measurement results.

4. The Hamiltonian we program

$$\frac{H(t)}{\hbar} = \underbrace{\sum_i \left[\frac{\Omega(t)}{2} \sigma_i^x - \Delta(t) \hat{n}_i \right]}_{\text{driving, programmed by the laser}} + \underbrace{\sum_{i<j} \frac{C_6}{\hbar R_{ij}^6} \hat{n}_i \hat{n}_j}_{\text{interaction, programmed by geometry}}$$

4.1 The interaction term

1. It is diagonal. The operator $\hat{n}_i \hat{n}_j$ is a projector: it costs energy if both atoms are excited, and does nothing otherwise.
2. It is always active and does not switch off when $\Omega = \Delta = 0$. A delay in a pulse sequence is not idle time but instead free evolution under the interactions.
3. It scales as R^{-6} , which is very steep (it becomes negligible very quickly). On a triangular lattice of spacing a , define the energy potential $U \equiv C_6/\hbar a^6$:

Position	Distance	Neighbours	Strength
Nearest	a	6	U
Next-nearest	$\sqrt{3}a$	6	$U/27$ (3.7 % of U)
Third	$2a$	6	$U/64$ (1.6 % of U)

A nearest-neighbour-only description is therefore a good approximation.

4.2 The spin mapping**The transverse-field Ising model**

$$\frac{H}{\hbar} = \Gamma \sum_i \sigma_i^x + J \sum_{\langle ij \rangle} \sigma_i^z \sigma_j^z - h \sum_i \sigma_i^z + \text{const},$$

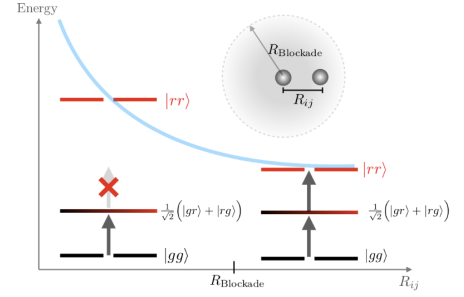
$$\Gamma = \frac{\Omega}{2}, \quad J = \frac{U}{4} > 0, \quad h = \frac{\Delta - \frac{z}{2}U}{2}.$$

- $J = U/4 > 0$ always holds. Rydberg repulsion is an unavoidable antiferromagnetic Ising coupling (a ferromagnet cannot be built this way). It is the machine's native ground state.
- $\Gamma = \Omega/2$ is the transverse field. The laser amplitude corresponds to the strength of the quantum-fluctuations.
- $h = (\Delta - \frac{z}{2}U)/2$ is the longitudinal field. It is the detuning, offset by the mean-field interaction energy.

5. The Rydberg blockade

When two atoms at distance R are driven resonantly from $|gg\rangle$, the interaction shifts the energy of the state $|rr\rangle$ upward by C_6/R^6 . If $\hbar\Omega \ll \frac{C_6}{R^6}$, the laser lacks the energy to reach $|rr\rangle$ and the excitation is blocked. Instead the pair is driven into the entangled singly-excited state $\frac{|gr\rangle+|rg\rangle}{\sqrt{2}}$. One excited atom therefore forbids its neighbours from being excited, and this is the microscopic origin of antiferromagnetic order: neighbours cannot both be up. The blockade radius is defined as,

$$R_b = \left(\frac{C_6}{\hbar\Omega} \right)^{1/6}.$$



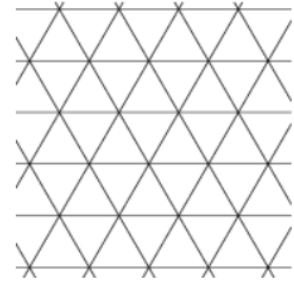
the transition to $|rr\rangle$ is suppressed when $R_{ij} \ll R_b$. arXiv:2104.15044v3, Quantum 6, 629 (2022).

6. The material: TmMgGaO₄

Step 1: Magnetic Tm ions form a triangular lattice

The magnetic degrees of freedom arise from Tm³⁺ ions. Within each magnetic layer the Tm sites form a two-dimensional triangular network, so each moment has $z = 6$ neighbours arranged around triangles, rather than along a chain or on a bipartite square lattice.

This single geometric fact is the source of everything interesting in the material, and it is the reason the standard textbook antiferromagnet does not apply.



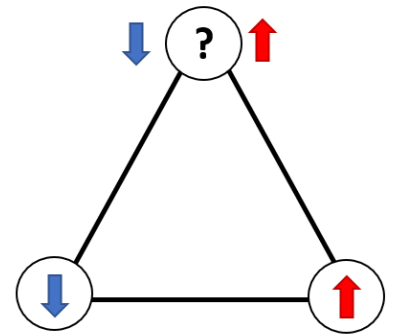
Triangular lattice of magnetic sites.

Step 2: The interactions are antiferromagnetic, and cannot all be satisfied

An antiferromagnetic interaction favours neighbouring moments pointing in opposite directions. For two moments this is trivial: $\uparrow\downarrow$. For three moments on a triangle it is impossible. Once two are fixed antiparallel, the third cannot satisfy both of its neighbours, and the result is geometrical frustration.

Two consequences, which we will make quantitative:

- There is no Néel state on a triangular lattice, and there is no checkerboard. The phrase antiferromagnetic state does not name the same object it names on a square lattice.
- Classically, at $h = 0$ and $\Gamma = 0$, the ground state is extensively degenerate. Any configuration with exactly one unsatisfied bond per triangle is a ground state, and there are exponentially many.



On a single triangle, not all three antiferromagnetic bonds can be satisfied at once.

Step 3: Strong anisotropy makes the moments Ising-like

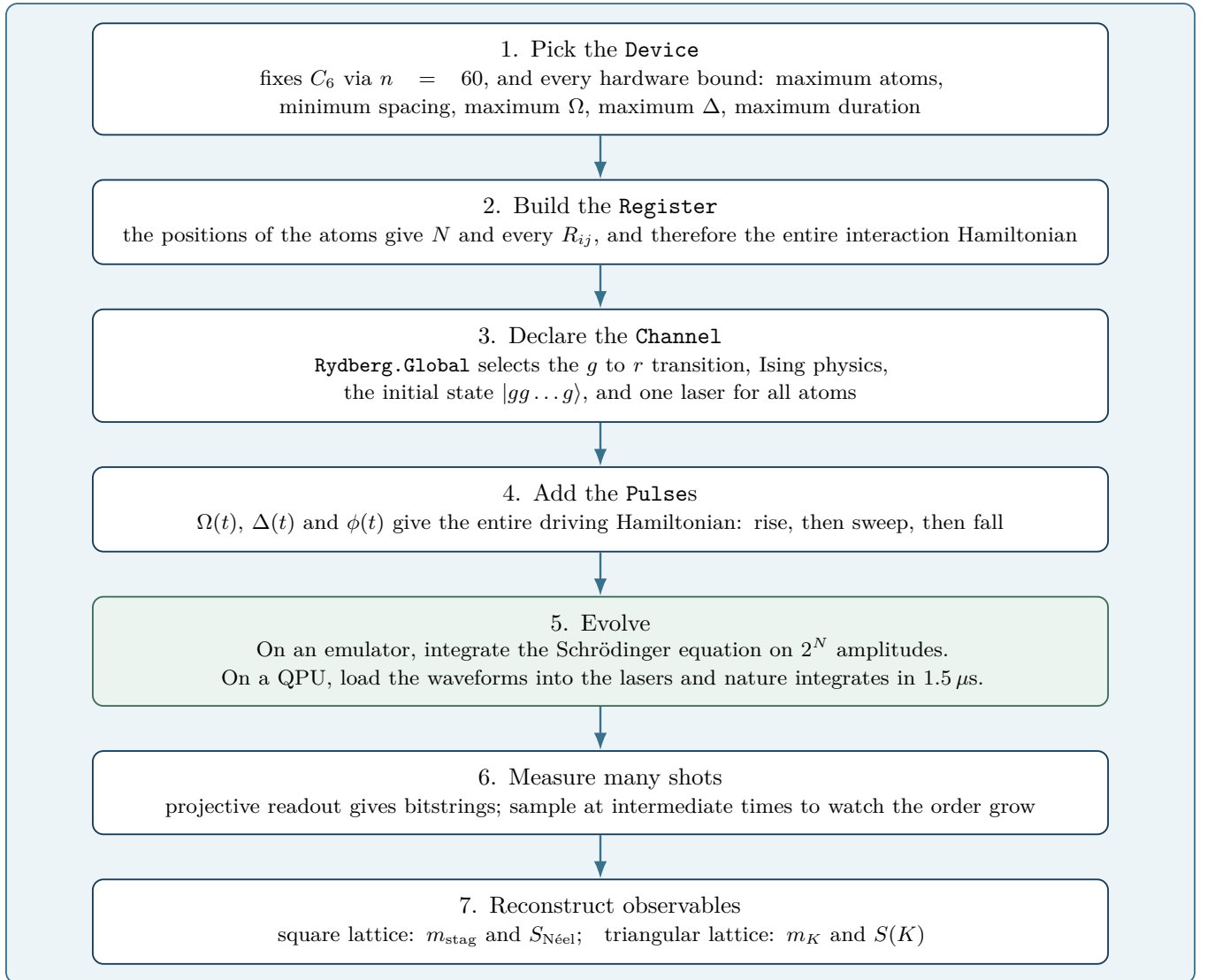
The crystalline electric field strongly favours moments along one axis, so the low-energy states of each Tm³⁺ ion reduce to two alternatives, $|\uparrow\rangle$ and $|\downarrow\rangle$. The material is therefore described by an effective

spin- $\frac{1}{2}$ Ising model, despite the complicated microscopic electronic structure. Crucially, the crystal-field splitting also supplies an intrinsic transverse field, so the material realises the transverse-field Ising antiferromagnet rather than the classical one.

Step 4: Encode the magnet as a neutral-atom spin model

One effective spin per site, and one atom per spin: $|g\rangle \equiv |\downarrow\rangle, |r\rangle \equiv |\uparrow\rangle$. Rydberg repulsion between neighbouring excited atoms is the neutral-atom realisation of antiferromagnetic Ising coupling, with $J = U/4$. The laser drive Ω plays the role of the crystal-field transverse field Γ , and the detuning Δ plays the role of the longitudinal field h .

7. Complete physics and code workflow



Where classical emulation stops working

The Hilbert space has dimension 2^N . Local state-vector emulation is comfortable up to roughly 15 or 18 atoms, painful at 20 and impossible at 30. Beyond that we need tensor networks, such as emu-mps, which reaches 60 to 100 atoms. But matrix product states are built for one dimension. Using them in two dimensions requires snaking the lattice into a chain, and the required bond dimension grows exponentially with the width of the snake. Frustrated two-dimensional systems near criticality are the worst case for this method, since they combine high entanglement with slow convergence.